# A Leakage-Aware Multimodal Crop-Storage Modeling Pipeline

*Simulation Audit and Public Tomato Scenario Study*

**UROP Project Report**

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# Acknowledgements:

I would like to express my sincere gratitude to [Current UROP Mentor Name] for their guidance and supervision during this UROP project. The opportunity to explore crop-storage monitoring through environmental readings and MQ sensor responses has helped me connect machine-learning methods with a practical postharvest question. I am grateful for the direction and feedback that helped me define a manageable scope, review the evidence carefully, and present the limitations of the work clearly.

I am also thankful to my two project teammates for their collaboration, discussions, and shared effort throughout the project. I extend my thanks to the UROP coordinators and the School of Engineering and Sciences for organizing this research opportunity and providing an academic setting in which to carry it out. Finally, I acknowledge the researchers and institutions whose public datasets, technical documentation, and postharvest guidance informed the literature review and data-source audit. Their work helped us distinguish what has been measured from what remains simulated in this study.


# 1. Introduction

Storage conditions can influence crop quality over time. This project builds a research workflow that combines environmental readings with responses from nine metal-oxide MQ channels: MQ2, MQ3, MQ4, MQ5, MQ6, MQ7, MQ8, MQ9, and MQ135. The intended tasks are to estimate current quality state and, when repeated independent inspections are available, risk by the next inspection interval.

The five-crop dataset covers wheat, maize, table potato, cured dry onion, and spinach. It contains 120,000 simulated rows: 500 starting lots, six storage paths per lot, and 40 readings per path. The readings, dates, labels, paths, and sensor values are generated. The rows are repeated simulator observations, not 120,000 real samples.

The goal of the current work is to verify the pipeline and evaluation design, not to claim validated crop accuracy. Simulator scores cannot establish how well the model will perform on real produce.

# 2. Literature Survey

The sources below support separate experiments and help define the future measurement plan. None provides a verified paired dataset containing the same crop lot, all requested sensors, and independent spoilage inspections.

| Study / dataset | What it contains | Use and limitation in this project |
|---|---|---|
| Tomato e-nose, Zenodo 15469472 [1] | 12 Rs/R0 gas-response channels, weight, colour, storage day and scenario. | Separate tomato storage-scenario experiment; no inspected spoilage target or matching AeroSense temperature/RH. |
| Nine-MQ recordings, Zenodo 17285312 [2] | Short recordings for the nine MQ channels; day-coded tomato and banana sessions. | Response-scale exploration only; no verified lot key, matched environment history, or independent inspection label. |
| Potato storage archive, Mendeley [3] | Temperature, RH, tuber-core temperature and CO2 histories across nodes and modes. | Useful environmental history; MQ and quality summaries lack a verified row-level lot/time join. |
| UC Davis postharvest guidance [4–6] | Crop-stage and intended-use storage guidance for tomato, potato and banana. | Informs simulated scenario centers, not spoilage labels or universal class thresholds. |

### Research gap

MQ sensors give cross-sensitive response patterns; raw readings are not exact gas concentrations without a defined measurement scale and calibration. The public records remain separate. Joining unrelated files would invent same-lot matches, so that was not done.

# 3. System Model

## 3.1 Five-crop dataset

The original CSV has 24,000 rows for each of five crops, for 120,000 rows total. This is 500 simulated starting lots × 6 storage paths per lot × 40 observations per path. Each crop contributes 100 starting lots, 600 paths, and 24,000 rows. All six paths from a starting lot remain together during splitting.

| Crop | Rows | Starting lots | Paths |
|---|---|---|---|
| Wheat | 24,000 | 100 | 600 |
| Maize | 24,000 | 100 | 600 |
| Table potato | 24,000 | 100 | 600 |
| Cured dry onion | 24,000 | 100 | 600 |
| Spinach | 24,000 | 100 | 600 |
| TOTAL | 120,000 | 500 | 3,000 |

## 3.2 Inputs and validation

Models compare environment-only, nine-MQ-only, and combined inputs. Predictors use information available by the current time. Status, inspection defects, failure targets, IDs, scenarios, fictional absolute dates, and future readings are excluded. A fixed split holds out 100 starting lots; five-fold grouped validation on 400 training lots tunes combined Random Forest and XGBoost. The starting-inspection field is evaluated separately because it must also be measured in a real deployment.

| Feature set | Inputs used at prediction time |
|---|---|
| Environment-only | Storage age, temperature, and relative humidity |
| MQ-only | MQ2, MQ3, MQ4, MQ5, MQ6, MQ7, MQ8, MQ9, and MQ135 response indices |
| Combined | The environment fields and all nine MQ response indices |

## 3.3 Survival and interface

The discrete-time hazard model uses readings at the start of each inspection interval and excludes post-failure readings. A lot still acceptable at monitoring end is censored, not assigned a made-up failure day. The output is risk by the next inspection, not an exact day-left estimate. The local browser app loads saved five-crop models; it is not connected to live hardware. V3 tomato/potato/banana models run separately through a command-line example.

## 3.4 Additional three-crop simulation

Edition 2 V3 contains 100 generated trajectories and eight readings per trajectory for tomato, table potato, and banana. Tomato and banana session medians anchor response scale only; potato MQ values use a generic synthetic scale. All environmental paths and three-state labels are simulated. This separate V3 experiment does not add measured lots to the five-crop dataset.

# 4. Results and Discussion

*All model scores in the following tables are simulator results, not validated crop accuracy. The five-crop test set contains repeated rows from 100 held-out starting lots.*

## 4.1 Five-crop status comparison

| Inputs / model | Macro F1 | Spoiled recall | Log loss | Brier |
|---|---|---|---|---|
| Majority baseline | 0.1794 | 0.0000 | 1.0962 | 0.6650 |
| Environment / XGBoost | 0.8360 | 0.8445 | 0.3921 | 0.2382 |
| Nine MQ / XGBoost | 0.9251 | 0.9526 | 0.1738 | 0.1084 |
| Combined / Random Forest | 0.9309 | 0.9587 | 0.1783 | 0.1055 |
| Combined / XGBoost | 0.9300 | 0.9562 | 0.1622 | 0.1004 |

XGBoost has better log loss and Brier probability scores; Random Forest has slightly higher macro F1 and spoiled recall. Paired whole-lot intervals for F1 and spoiled recall include zero, so neither model clearly wins those measures. XGBoost is a reasonable probability-focused simulator model, while Random Forest remains a close benchmark.

## 4.2 Per-crop XGBoost results

| Crop | Macro F1 | Spoiled recall |
|---|---|---|
| Wheat | 0.9185 | 0.9536 |
| Maize | 0.9232 | 0.9394 |
| Table potato | 0.9400 | 0.9621 |
| Cured dry onion | 0.9165 | 0.9391 |
| Spinach | 0.9506 | 0.9718 |

## 4.3 Three-crop V3 and time-risk results

| V3 crop | RF F1 / high-risk recall | XGBoost F1 / high-risk recall |
|---|---|---|
| Tomato | 0.787 / 0.630 | 0.746 / 0.667 |
| Potato | 0.614 / 0.400 | 0.580 / 0.320 |
| Banana | 0.687 / 0.633 | 0.745 / 0.733 |

In the five-crop simulation, next-interval XGBoost hazard Brier was 0.01545 (AUC 0.98036); logistic Brier was 0.05504 (AUC 0.98297). AUC is not days-left accuracy. Separately, real tomato data labels storage scenario, not spoilage: XGBoost macro F1 was 0.498 under leave-one-day-out validation and 0.278 on nine tracked tomatoes. This small transfer check is weak.

# 5. Concluding Remarks

A runnable workflow now covers data review, grouped model comparison, saved-model predictions, and interval-based risk. Thirty focused checks pass for grouped splitting, leakage exclusions, censoring, prediction inputs, and V3 sample-time keys. High simulator scores show that the code can learn generated patterns; they do not show that its labels or sensor patterns match real crop quality.

# 6. Future Work

For real validation, record temperature/RH and all nine MQ responses for the same identified lot at matched times, with sensor calibration and warm-up details. Independent assessors should apply a written crop-specific rubric: acceptable means sound and marketable; warning means early deterioration or limited defects needing recheck; high risk means severe decay or nonmarketable condition. Record inspection evidence and time. Store the last acceptable and first unacceptable inspection as an interval; record monitoring end as censoring if no failure occurs. Hold out complete lots and preferably a later harvest or device. Compare real-only training with real-plus-simulation on untouched real lots.

Future temperature/RH what-if results require explicit future MQ values and are conditional projections; they do not prove cooling causes longer shelf life.

### Minimum paired-data record

| Record | Fields to capture |
|---|---|
| Sensor observation | Study and lot IDs; crop and stage; UTC timestamp and storage age; temperature and RH with units; all nine MQ responses; device, calibration, and warm-up details. |
| Independent inspection | Same lot/sample ID; inspection time; assessor and method; acceptable/warning/high-risk state; crop-specific defect evidence. |
| Follow-up | Last acceptable and first unacceptable inspection times, or monitoring end if no failure; keep all readings from one lot in one split. |

### References


[1] Zenodo. Tomato e-nose and laboratory data, record 15469472. https://zenodo.org/records/15469472

[2] Zenodo. Nine-MQ food freshness recordings, record 17285312. https://zenodo.org/records/17285312

[3] Mendeley Data. Potato storage-condition histories, sbwxmtpj47, version 1. https://data.mendeley.com/datasets/sbwxmtpj47/1

[4] UC Davis Postharvest Center. Tomato Produce Facts. https://postharvest.ucdavis.edu/produce-facts-sheets/tomato

[5] UC Davis Postharvest Center. Potato Produce Facts. https://postharvest.ucdavis.edu/produce-facts-sheets/potato

[6] UC Davis Postharvest Center. Banana Produce Facts. https://postharvest.ucdavis.edu/produce-facts-sheets/banana

[7] XGBoost Developers. AFT Survival Analysis. https://xgboost.readthedocs.io/en/stable/tutorials/aft_survival_analysis.html